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title: "Speed sim"
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The rules of Speed

Speed is a game played by with a number of stacks (usually 2) of cards with one card each to start, then the rest of the cards dealt evenly to each player. Each player picks up 5 cards, and they can play the card on any of the stacks if the stacked card is one above or below theirs. There is no turn order, everyone plays as fast as possible. The first one to empty their deck wins. (an ace can be played on a king and vice versa)

I want to know the probability of getting stuck i.e. no player is able to play any of their cards. Normally, two reserve piles of 3 cards each are dealt during setup to refresh the top card when both players get stuck.

Variables

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```{r}
players_num = 2 # number of players
stack_num = 2 # number of stacks to play cards on
reserve_size = 3 # number of cards in each reserve stack (there is one reserve for every play
stack)
hand_size = 5 # number of playable cards at any one time
suits = 4 # how many suits the deck has
numbers = 13 # how many cards are in each suit
```

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Functions

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```{r}
make_decks <- function(n, size) { # returns list of n decks of size cards pulled from the global
"deck" variable
 cards <- deck[1:(n*size)]
 decks <- split(cards, 1:n)
 deck <- deck[-(1:(n*size))]
 return(decks)
}
```

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Setup

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Create a list of 52 cards in random order
Stacks is a list of vectors, one for each playable stack on the table
Reserves is a list of vectors, one for each reserve pile
Hands is a list of vectors, one for each player's deck
```{r}
setup <- function() {
 suit <- 1:numbers
 deck <- sample(rep(suit, suits)) #sample randomly permutes the deck
 # Pick starting cards
 stacks <- make_decks(stack_num,1)

 # Make reserve decks
 reserves <- unlist(make_decks(1,reserve_size * stack_num))

 # Deal remaining cards to each player
 hands <- make_decks(players_num, length(deck)/players_num)
}
```

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## Play
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The players alternate playing the first of their first five cards on a stack that they can, and continue alternating until they get stuck.

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``` {r}
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```
play_card <- function (hand, debug = 0) { # DEBUG FUNCTION INCLUDING PRINT STATEMENTS hand is a
number representing the player that is playing the card
 playing_hand <- as.vector(na.omit(hands[[hand]][1:hand_size])) # take the first 5 cards from the
hand playing the card
 if (debug >= 1) {print(c("Hand: ", hand, "|", playing_hand))}
 stack_index <- 0
 played <- FALSE
 for (stack in stacks) { # iterate over each stack
 stack_index <- stack_index + 1
 if (debug >= 2) {print(c("Stack ", stack_index, stack[[1]]))}
 card_index <- 0
 for (card_to_play in playing_hand) { # iterate over each card
 card_index <- card_index + 1 # increase the index of the card
 if (((card_to_play%numbers) + 1) == stack[[1]] || card_to_play == ((stack[[1]]%numbers) +
1)) { #checks if the card can be played on the stack
 if (debug >= 1) {print(c(card_to_play, "playable!"))}
 stack <- c(card_to_play, stack) # add playable card to the stack
 hands[[hand]] <- hands[[hand]][-card_index] # take the card out of the global hand
variable
 stacks[[stack_index]] <- stack # replace the stack in the global stacks variable
 return(card_to_play)
 } else if (debug >= 2) {print(c(card_to_play, "Not playable!"))}
 }
 }
 return(0)
}
```
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## Game function
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A function containing the whole game, returning true if all cards get played and false if neither player can play any cards.

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``` {r}
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play_speed <- function (n, debug = 0) { # attempts to play n rounds in a game of speed
 setup()
 for (round in 1:n) {
 if (debug >= 1) {print(round)}
 remainder <- length(as.vector(unlist(hands)))
 card_sum <- 0
 for (player in 1:players_num) {
 card_sum <- card_sum + play_card(player, debug = debug)
 if (debug >= 1) {print(c("sum of cards played", card_sum))}
 }
 if (card_sum == 0 && length(reserves)>0) { # put top card of reserve onto stacks
 stack_index <- 0
 for (stack in stacks) {
 stack_index <- stack_index + 1
 stacks[[stack_index]] <- c(reserves[[1]], stack) # add card to the stack
 reserves <- reserves[-1] # take the card off the reserves stack
 }
 } else if (card_sum == 0 && length(reserves) == 0 && remainder > 0) { # the game failed to
finish
 return(FALSE)
 } else if (remainder == 0) { # the game did finish
 return(TRUE)
 }
 }
}
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## Replication
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Replicate the simulation n times, find the probability that the game finishes
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```{r}
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n = 5000
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mean(replicate(n, play_speed(50)))
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